NATIONAL UNIVERSITY OF SCIENCES & TECHNOLOGY

Machine Learning (Transformer Based Grammar Aid)

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GENERAL INFORMATION (IN WORKFLOW ORDER)

1. **Gradio:** A Python library to create simple web interfaces for machine learning models, enabling interactive demos. Provides a user-friendly GUI where users input sentences, see corrected output, and get grammatical explanations.

2. **Hugging Face Transformers:** A library providing pre-trained Transformer models for NLP tasks like text generation, summarization, translation, and grammar correction. We use the prithivida/grammar_error_correcter_v1 model to automatically correct grammatical errors in user-input sentences.

3. **Tokenizer:** A tokenizer converts human-readable text into numeric tokens that a model can understand, and vice versa. AutoTokenizer encodes the input sentence into tokens for the model and decodes the model's output tokens back to readable text that is displayed on User Interface.

4. **Natural Language Processing (NLP):** A field of computer science and AI focused on processing and understanding human language. NLP as a general concept is overarching, not a specific step in the workflow. Think of it like the umbrella under which both spaCy and the Transformer model operate.

5. **Prithivida/grammar_error_correcter_v1:** A pre-trained Transformer-based model available on Hugging Face. It is specifically designed for grammar error correction in English. Application in Program: This Transformer model corrects the sentence. It takes your input and outputs a grammatically improved version.

6. **SpaCy:** A popular NLP library for tokenization, part-of-speech (POS) tagging, lemmatization, and syntactic analysis. Analyzes the original and corrected sentences to identify grammatical structures and changes.

   Example output for "She goes to school":

   - Tokens: ["She", "goes", "to", "school"]
   - POS tags: PRON, VERB, ADP, NOUN
   - Lemmas: "she", "go", "to", "school"
   - Dependencies: subject → verb → object

References: The following resources provided foundational knowledge and practical guidance essential for understanding and implementing Transformer based grammar correction systems.

Sequence Models | DeepLearning.AI

Gen AI Foundational Models for NLP & Language Understanding | IBM

Generative AI and LLMs: Architecture and Data Preparation | IBM

Generative AI Language Modeling with Transformers | IBM

Generative AI Engineering and Fine-Tuning Transformers | IBM

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